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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Examination December 2022 (2019 scheme)



Course Code: EET401

Course Name: ADVANCED CONTROL SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- 1 What do you mean by (i) state variables (ii) phase variables and (iii) state space? (3)
 - 2 A series RLC circuit is excited by a voltage source, $v(t)$ volts and the output is measured across the resistor. Derive the state model of the electrical system. (3)
 - 3 State and prove any three properties of state transition matrix. (3)
 - 4 The state space representation and transfer function of a system are given below. Find the value of 'm'. (3)
- $$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
- $$\frac{Y(s)}{U(s)} = \frac{2s+2}{(s+1)(s+4)}$$
- 5 With necessary equations, explain duality principle. (3)
 - 6 Explain the Gilbert's method to test controllability of the n^{th} order system represented by the state model given below. Assume that the eigen values are distinct. (3)
- $$\dot{X} = AX + BU \quad Y = CX + DU$$
- 7 Write any three characteristics of non-linear systems. (3)
 - 8 Explain the following (i) jump resonance (ii) stable limit cycle and (iii) unstable limit cycle? (3)
 - 9 Define (i) singular point (ii) vortex point and (iii) saddle point. (3)
 - 10 State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each. (3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) From basics, derive the state model of an armature controlled dc motor. (7)
- b) Differential equation of the system is given by, (7)

$$\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0$$

where, y is the output and u is the input. Derive the state model of the system using phase variables.

OR

- 12 a) What is similarity transformation? How similarity transformation is helpful for state space analysis? (5)
- b) Obtain the diagonal canonical form of the state model given below. (9)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u \quad [y] = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Module II

- 13 A discrete time system is given by, (14)
- $$y(k+2) + 5y(k+1) + 6y(k) = u(k)$$

where, y is the output and u is the input. Assume zero initial conditions and the sampling period as 1 second. (i) Derive the state model of the system in Jordan canonical form (ii) Draw the block diagram of the state model and (iii) Compute the state transition matrix.

OR

- 14 a) State equation of a system is given below. Compute the state transition matrix using Laplace transform method. Also, compute the solution of the given state equation if the initial state vector is $X(0)$. (7)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- b) Find the state transition matrix using Cayley-Hamilton theorem for the system matrix given below. (7)

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Module III

- 15 a) Using Kalman's method, check the observability of the system given below. (6)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [y] = [40 \quad 10 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- b) Design the observer feed back gain matrix for the desired eigen values at $-4, -3 \pm j1$ for the state model of the system given below. (8)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

OR

- 16 a) Using PBH test, check the controllability of the system given below. (6)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- b) Design a feed back controller with state feed back so that the closed loop poles are at $-2, -1 \pm j1$. The state equation of the original systems is given below. (8)

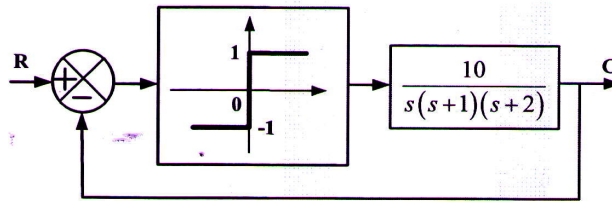
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$$

Module IV

- 17 a) Derive the describing function of ideal relay non-linearity given below. (6)

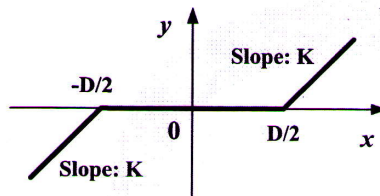


- b) Determine the frequency and nature of the limit cycle for the unity feed back system given below. (8)

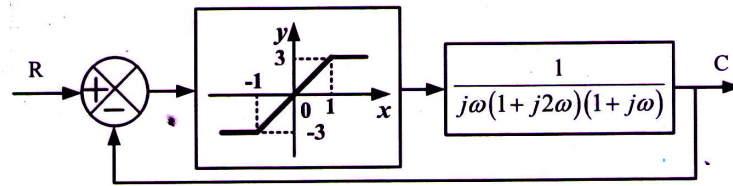


OR

- 18 a) Derive the describing function of dead-zone non-linearity given below. (6)



- b) Determine the frequency and nature of the limit cycle for the unity feed back system given below. (8)

**Module V**

- 19 a) What do you mean by phase trajectory? Explain how to draw the phase trajectory using isocline method. (7)
- b) Compute the Lyapunov function, $V(X)$ for which the system given below is asymptotically stable. (7)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

OR

- 20 a) Construct the phase trajectory of the second order system given below using isocline method. (7)

$$\dot{x}_2 + (2\xi\omega_n)x_2 + (\omega_n^2)x_1 = 0$$

$$X(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} \quad \xi = 0.15 \quad \omega_n = 1 \text{ rad/s}$$

where x_1 and x_2 are the state variables.

- b) Explain Lyapunov function? Using Lyapunov function, $V(X)$, state the condition/s for the system to be (i) stable (ii) asymptotically stable in the large and (iii) unstable. (7)

Consider the system described by,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - x_2^3$$

Check the stability of the system for the Lyapunov function, $V(X) = x_1^2 + x_2^2$.
